

Changes to the ship workbench:



Loading a geometry

In addition to the four known ship types, you can load your own ship or geometry.

The following options are available for this purpose

- * Use sample ship

- * Import a geometry as:

STL file, the shape must be closed, i.e., including deck and transom. If there is no closed shape, the STL file cannot be converted into a solid.

IGES or STEP files are usually solids.

GF / GF1 files; these point specifications are combined into frames and must be converted manually into solids. It is recommended to create a closed mesh using lofts and then convert it into a solid in the part workbench

- * Own geometry; this must be a solid and can then be saved and edited as a ship instance.

You can choose a unit of measurement during import or automatic conversion of the measurements. FreeCAD calculates internally in mm.

The structure is placed so that the coordinate origin is at $L/2$. This is rather unusual in shipbuilding; normally, the coordinate origin is at the rear plumb line in Europe and at the front plumb line in the USA. Just a note, as this coordinate origin is rather unusual. This must be taken into account, especially for tank positions and weight positions.



Ships-area and Hydrostatic, as well as Create New tank and Tank capacity curve are unchanged from the original version.

The major difference to the previous version is the weight calculation. Until now, the data was stored exclusively in the ship instance and was somewhat difficult to influence or edit.



Now, a central load condition spreadsheet has been created in which you can fill the tanks you have created, listing all weights and cargo. There are three different groups:

Tanks, with the corresponding density of the liquid, percentage fill, and free liquid surfaces



Weights: here, primarily unchanging weights are assumed, such as: light ship weight, crew & effects, provisions, stores. The create new ship weight button is intended for these weights.



Once tanks and weights have been defined, a spreadsheet with the weights can be created using the create new loading condition button. The fill levels of the tanks can be entered as a percentage in this spreadsheet.



In addition, cargo can be imported. A packing list is best suited for this purpose. Clicking on the spreadsheet icon opens the import function for a packing list. On the start page of the import function, you will be asked for the packing list file. If there are several sheets, select the desired sheet. Under the Columns tab, assign the columns for name, description, weight, length, width, and height. Under the Settings tab, assign the units m, mm, kg, t, etc. Then, click on the Start Import button to start the import

Cuboids with specified length/width/height are formed, the labeling is engraved, and at the same time these weights and positions (assumed COG L/2, W/2, H/2) are entered into the load condition.

It is possible to move the cuboids to where you want them, and a stowage plan can be generated in this way.



Various columns in the load condition are still empty. The next step is the initial calculation of the given load condition. To do this, press the Calculate Load Case button, and the total weight and moments are then calculated.



With the Sink and Trim vessels button, equilibrium will be calculated with GM of load condition. The first time you call this up, a spreadsheet with different drafts and trims is saved, so that when you call it up again, the iterations to determine the ship's floating position are much faster.



The GZ curve shows the current stability and relates it to SOLAS requirements to determine whether the ship meets the stability criteria. Min. GM due to a leak calculation is not taken into account here.

The areas for resistance calculation and RAOs are unchanged, although I am having difficulty starting Capytaine via FreeCAD.